Name: _____

TOPIC TEST 2

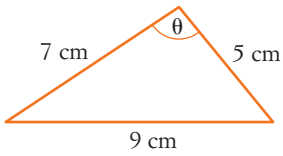
Chapter 3: Trigonometry

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 10 short answer questions (46 marks)
- Part C: 2 analysis questions (20 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Which statement is true?



A $\cos(\theta) = \dfrac{5^2 + 9^2 - 7^2}{2 \times 5 \times 9}$

B $\cos(\theta) = \dfrac{5^2 + 7^2 - 9^2}{5 \times 7}$

C $\cos(\theta) = \dfrac{9 + 7 - 5}{2 \times 9 \times 7}$

D $\cos(\theta) = \dfrac{5^2 + 7^2 - 9^2}{2 \times 5 \times 7}$

E $\cos(\theta) = \dfrac{5^2 + 7^2 - 9^2}{5 \times 7}$

[1 mark]

© Cengage Learning Australia Pty Ltd 2013

2 $\tan\left(\frac{4\pi}{3}\right) =$

A $\frac{1}{\sqrt{3}}$

B $-\sqrt{3}$

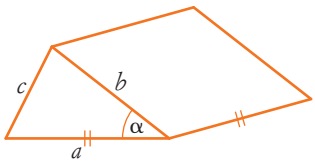
C $\sqrt{3}$

D $-\frac{1}{\sqrt{3}}$

E 1

[1 mark]

3 What is the volume of the figure below?



A $V = \frac{1}{2}a^2b\sin(\alpha)$

B $V = \frac{1}{2}abc\sin(\alpha)$

C $V = \frac{1}{2}abc\cos(\alpha)$

D $V = \frac{1}{2}a^2c\sin(\alpha)$

E $V = \frac{1}{2}a^2b\cos(\alpha)$

[1 mark]

4 Which statement is NOT true?

A $\cos(360° - \theta) = -\cos(\theta)$

B $\sin(180° - \theta) = \sin(\theta)$

C $\tan(360° - \theta) = -\tan(\theta)$

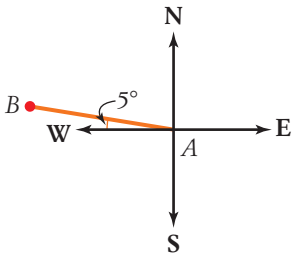
D $\cos(-\theta) = \cos(\theta)$

E $\sin(180° + \theta) = -\sin(\theta)$

[1 mark]

© Cengage Learning Australia Pty Ltd 2013

5 What is the true bearing of B from A in the figure below?



A 005°

B 085°

C 265°

D 185°

E 275°

[1 mark]

Part B

10 short answer questions
46 marks
Show your working where appropriate.

1 Convert the following to degrees.

a $\frac{\pi}{2}$

b $\frac{2\pi}{3}$

c $\frac{5\pi}{6}$

d $\frac{7\pi}{4}$

e 5π

[5 marks]

2 Convert the following to radians, correct to one decimal place.

a 74°

b 29°

c 108°

d 41.6°

e 84° 12′

[5 marks]

3 A road has an angle of inclination of 8° 52′. What is its gradient?

[1 mark]

4 Evaluate the following, correct to two decimal places.

a $\cos\left(\frac{\pi}{7}\right)$ _____

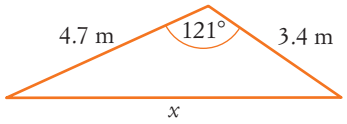
b tan (0.9) _____ **[2 marks]**

5 A fountain is 21.5 m out from the base of a building. Amie is standing at a window in the building that is 7.3 m up from the ground. What is the angle of depression from Amie down to the fountain, correct to one decimal place?

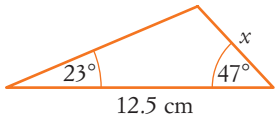
[3 marks]

6 Evaluate x correct to one decimal place.

a

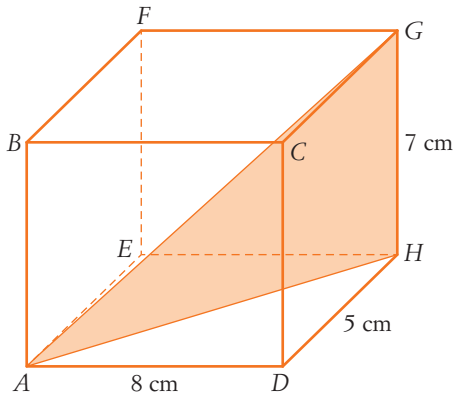


b



[7 marks]

7 A box with sides 8 cm, 7 cm and 5 cm has a piece of cardboard inserted inside it, as shown in the diagram below. Find the size of angle *GAH*, correct to the nearest degree.



[4 marks]

8 A sector is cut off by an angle of $\frac{3\pi}{7}$ subtended at the centre of a circle with radius 8.6 cm.

a Find the length of the arc cut off by this angle.

[2 marks]

b Find the area of the sector.

[2 marks]

[4 marks]

9 State which quadrant each of the following angles are in.

a $-50°$ _____

b $500°$ _____

c $-285°$ _____

d $\frac{7\pi}{6}$ _____

e $\frac{3\pi}{5}$ _____

f $-\frac{13\pi}{4}$ _____

[6 marks]

10 The circumference of a circle is 20 cm. Find the angle subtended at the centre if it cuts off a sector with an area of 15 cm^2.

[5 marks]

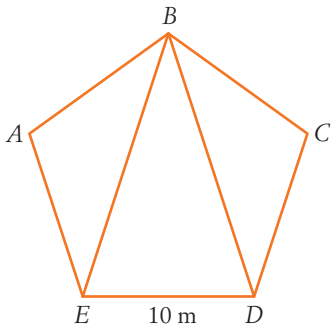
Part C

2 analysis questions
20 marks
Show your working where appropriate.

1 A plane leaves Brisbane and flies for 113 km on a bearing of 058°. It then turns and flies on a second bearing for 154 km until it is 210 km due east of Brisbane. What is the second bearing, to the nearest degree?

[7 marks]

2 The diagram below shows a regular pentagon with sides of 10 m.



a Use the sum of angles in a triangle to find the angle sum of any pentagon.

b Find the size of each angle inside a regular pentagon.

c Find the length of *BE* in the pentagon above.

d Find the area of the pentagon.

[13 marks]
Total marks: 71

© Cengage Learning Australia Pty Ltd 2013

Answers

Part A **[1 mark each]**

1 Substituting into the cosine rule:

$$\cos(C) = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos(\theta) = \frac{5^2 + 7^2 - 9^2}{2 \times 5 \times 7}$$

$\therefore$ **D** **[1 mark]**

2 $\frac{4\pi}{3} = \frac{3\pi}{3} + \frac{\pi}{3}$

$= \pi + \frac{\pi}{3}$

$\tan\left(\frac{4\pi}{3}\right) = \tan\left(\pi + \frac{\pi}{3}\right)$

$= \tan\left(\frac{\pi}{3}\right)$ (3rd quadrant)

$= \sqrt{3}$

$\therefore$ **C** **[1 mark]**

3 $V = Ah$ where $h = a$ and

$A = \frac{1}{2}ab\sin(C)$

$V = \frac{1}{2}ab\sin(\alpha) \times a$

$= \frac{1}{2}a^2b\sin(\alpha)$

$\therefore$ **A** **[1 mark]**

4 **A** $\cos(360° - \theta) = -\cos(\theta)$ is false (4th quadrant − cos is positive)

B $\sin(180° - \theta) = \sin(\theta)$ is true (2nd quadrant − sin is positive)

C $\tan(360° - \theta) = -\tan(\theta)$ is true (4th quadrant − tan is negative)

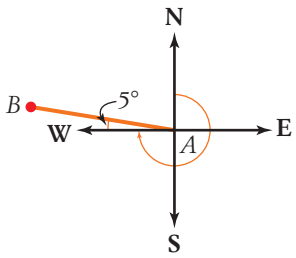
D $\cos(-\theta) = \cos(\theta)$ is true (4th quadrant − cos is positive)

E $\sin(180° + \theta) = -\sin(\theta)$ is true (3rd quadrant − sin is negative)

$\therefore$ **A** **[1 mark]**

© Cengage Learning Australia Pty Ltd 2013

5



True bearings are measured clockwise from north.

The bearing is $270° - 5°$ or $265°$.

$\therefore$ **C** **[1 mark]**

Part B

1 To convert to degrees, multiply by $\frac{180°}{\pi}$ or simply replace π by $180°$.

a $\frac{\pi}{2} \times \frac{180°}{\pi} = 90°$ or $\frac{\pi}{2} = \frac{180°}{2} = 90°$ **[1 mark]**

b $\frac{2\pi}{3} \times \frac{180°}{\pi} = 120°$ or $\frac{2\pi}{3} = \frac{2 \times 180°}{3} = 120°$ **[1 mark]**

c $\frac{5\pi}{6} \times \frac{180°}{\pi} = 150°$ or $\frac{5\pi}{6} = \frac{5 \times 180°}{6} = 150°$ **[1 mark]**

d $\frac{7\pi}{4} \times \frac{180°}{\pi} = 315°$ or $\frac{7\pi}{4} = \frac{7 \times 180°}{4} = 315°$ **[1 mark]**

e $5\pi \times \frac{180°}{\pi} = 900°$ or $5 \times 180° = 900°$ **[1 mark]**

2 To convert to radians, multiply by $\frac{\pi}{180°}$.

a $74° \times \frac{\pi}{180°} = 1.3$ **[1 mark]**

b $29° \times \frac{\pi}{180°} = 0.5$ **[1 mark]**

c $108° \times \frac{\pi}{180°} = 1.9$ **[1 mark]**

d $41.6° \times \frac{\pi}{180°} = 0.7$ **[1 mark]**

e $84° 12' \times \frac{\pi}{180°} = 1.5$ **[1 mark]**

3 Gradient $= \tan(8° 52')) = 0.16$ **[1 mark]**

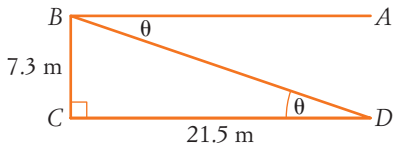
© Cengage Learning Australia Pty Ltd 2013

4 Set the calculator on radian mode.

a $\cos\left(\frac{\pi}{7}\right) = 0.90$ **[1 mark]**

b $\tan(0.9) = 1.26$ **[1 mark]**

5 Draw a diagram showing the information.



[1 mark]

The angle of depression is $\angle ABD$.

$\angle ABD = \angle BDC$ since $AB \| DC$

$\tan(\theta) = \frac{7.3}{21.5}$ **[1 mark]**

$\theta = 18.8°$

So the angle of depression is 18.8°. **[1 mark]**

6 a Substitute into the cosine rule:

$c^2 = a^2 + b^2 - 2ab\cos(C)$

$x^2 = 4.7^2 + 3.4^2 - 2 \times 4.7 \times 3.4 \times \cos(121°)$ **[1 mark]**

$= 50.110\,616\,87\ldots$ **[1 mark]**

$x = \sqrt{50.110\,616\,87\ldots}$

≈ 7.1

So $x = 7.1$ m **[1 mark]**

b The 3rd angle in the triangle is $180° - (23° + 47°)$ or 110°. **[1 mark]**

Substitute into the sine rule:

$\frac{a}{\sin(A)} = \frac{b}{\sin(B)}$

$\frac{x}{\sin(23°)} = \frac{12.5}{\sin(110°)}$ **[1 mark]**

$x = \frac{12.5\sin(23°)}{\sin(110°)}$ **[1 mark]**

≈ 5.2

So $x = 5.2$ cm **[1 mark]**

© Cengage Learning Australia Pty Ltd 2013

7 Use Pythagoras' theorem to find AH:

$c^2 = a^2 + b^2$

$AH^2 = 8^2 + 5^2$ **[1 mark]**

$= 89$

$AH = \sqrt{89}$ **[1 mark]**

Find angle GAH.

$\tan(\angle GAH) = \frac{7}{\sqrt{89}}$ **[1 mark]**

$\angle GAH = 37°$ **[1 mark]**

8 a Substitute into $l = r\theta$:

$l = 8.6 \times \frac{3\pi}{7}$ **[1 mark]**

$= 11.6$

So the arc length is 11.6 cm. **[1 mark]**

b Substitute into $A = \frac{1}{2}r^2\theta$:

$A = \frac{1}{2} \times 8.6^2 \times \frac{3\pi}{7}$ **[1 mark]**

$= 49.8$

So the area of the sector is 49.8 cm^2. **[1 mark]**

9 a $-50°$ is between $0°$ and $-90°$ so it is in the 4th quadrant (turn anticlockwise $50°$) **[1 mark]**

b $500° - 360° = 140°$ (turn around the circle more than once)

$140°$ is between $90°$ and $180°$ so it is in the 2nd quadrant. **[1 mark]**

c $-285°$ is between $-270°$ and $-360°$ so is in the 1st quadrant. **[1 mark]**

d $\frac{7\pi}{6} = \frac{6\pi}{6} + \frac{\pi}{6} = \pi + \frac{\pi}{6}$ which is in the 3rd quadrant. **[1 mark]**

e $\frac{3\pi}{5} = \frac{5\pi}{5} - \frac{2\pi}{6} = \pi - \frac{\pi}{3}$ which is in the 2nd quadrant. **[1 mark]**

f $-\frac{13\pi}{4} = -\frac{12\pi}{4} - \frac{\pi}{4} = -\left(3\pi + \frac{\pi}{4}\right) = -\frac{5\pi}{4} = -\left(\pi + \frac{\pi}{4}\right)$ which is in the 2nd quadrant. **[1 mark]**

10 Substitute in $C = 2\pi r$

$20 = 2\pi r$ **[1 mark]**

$\frac{20}{2\pi} = r$

$\frac{10}{\pi} = r$ **[1 mark]**

Substitute into $A = \frac{1}{2}r^2\theta$

$15 = \frac{1}{2} \times \left(\frac{10}{\pi}\right)^2 \theta$ **[1 mark]**

$= 5.066\,\theta$

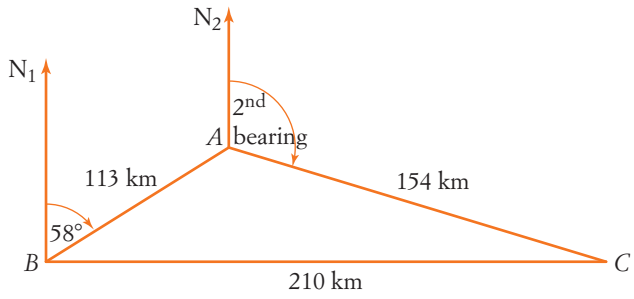
$\frac{15}{5.066} = \theta$ **[1 mark]**

$2.96 = \theta$

So the angle is 2.96. **[1 mark]**

Part C

1 Draw a diagram to show the information.



[1 mark]

Find $\angle BAC$:

Substitute into the cosine rule:

$$\cos(C) = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos(\angle BAC) = \frac{113^2 + 154^2 - 210^2}{2 \times 113 \times 154}$$ **[1 mark]**

$= -0.21879669$ **[1 mark]**

$\angle BAC = 102.638\ldots°$ **[1 mark]**

$\angle BAN_2 = 180° - 58°$

$= 122°$ **[1 mark]**

$\angle CAN_2 = 360° - (122° + 102.638\ldots°)$ **[1 mark]**

$= 135.361\ldots°$

So the second bearing is about 135°. **[1 mark]**

2 a A pentagon is made up of 3 triangles.

Angle sum $= 3 \times 180° = 540°$ **[1 mark]**

b There are 5 equal angles in a regular pentagon.

$\frac{540°}{5} = 108°$

So each angle is 108°. **[1 mark]**

c Substitute into the cosine rule:

$c^2 = a^2 + b^2 - 2ab\cos(C)$

$BE^2 = 10^2 + 10^2 - 2 \times 10 \times 10 \times \cos(108°)$ **[1 mark]**

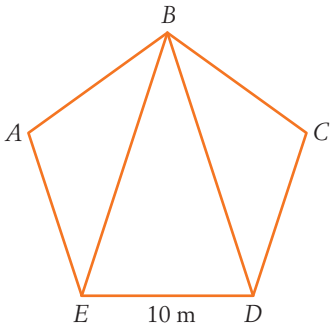
$= 261.8033989$ **[1 mark]**

$BE = \sqrt{261.8033989}$

$= 16.180\ldots$

So the length of BE is about 16.2 m. **[1 mark]**

d



Area of $\triangle ABE$:

Substitute into area formula $A = \frac{1}{2}ab\sin(C)$:

$A = \frac{1}{2} \times 10 \times 10 \times \sin(108°)$ **[1 mark]**

$= 47.552\ldots$

Area of $\triangle CBD$ is also 47.552… (both triangles are identical) **[1 mark]**

In $\triangle ABE$:

$\angle ABE = \frac{180° - 108°}{2}$ (base angles of an isosceles triangle)

$= 36°$

Similarly, $\angle CBD = 36°$

So $\angle EBD = 108° - 2 \times 36°$

$= 36°$ **[2 marks]**

Area of $\triangle EBD$:

$A = \frac{1}{2} \times 16.180\ldots \times 16.180\ldots \times \sin(36°)$ **[1 mark]**

$= 76.942\ldots$

The area of $\triangle EBD$ is $76.942\ldots$ m^2. **[1 mark]**

Total area $= 47.552\ldots + 47.552\ldots + 76.942\ldots$ **[1 mark]**

$= 172.047\ldots$

So the area of the pentagon is about 172.0 m^2. **[1 mark]**

© Cengage Learning Australia Pty Ltd 2013